

Lab4_controller_rviz

random_pose_node.py

Publisher

/target [geometry_msgs/msg/PoseStamped]

Subscriber

Service Server

/get_target_pose [lab4_controller_rviz/srv/GetTargetPose]

Service Client

teleop_jog_keyboard.py

Publisher

/cmd_vel [geometry_msgs/msg/Twist]
/velocity_frame [std_msgs/msg/String]

Subscriber

Service Server

Service Client

end_effector_publisher.py

Publisher

/end_effector [geometry_msgs/msg/PoseStamped]

Subscriber

/joint_states [sensor_msgs/msg/JointState]

Service Server

Service Client

controller_Node_rviz.py

Publisher

/joint_commands [std_msgs/msg/Float64MultiArray]
/singularity_warning [std_msgs/msg/String]

Subscriber

/joint_states [sensor_msgs/msg/JointState]
/cmd_vel [geometry_msgs/msg/Twist]
/target [geometry_msgs/msg/PoseStamped]
/velocity_frame [std_msgs/msg/String]

Service Server

/compute_ik [lab4_controller_rviz/srv/ComputeIK]
/set_control_mode [lab4_controller_rviz/srv/SetControlMode]

Service Client

/get_target_pose [lab4_controller_rviz/srv/GetTargetPose]

workspace_analyzer_node

Publisher

/workspace_points [sensor_msgs/msg/PointCloud2]

Subscriber

Service Server

Service Client

mock_robot_node.py

Publisher

/joint_states [sensor_msgs/msg/JointState]

Subscriber

/joint_commands [std_msgs/msg/Float64MultiArray]

Service Server

Service Client

robot_state_publisher

Publisher

/robot_description [std_msgs/msg/String]

Subscriber

/joint_states [sensor_msgs/msg/JointState]

Service Server

Service Client

Terminal

- Publisher
- Subscriber
- Service Server
- Service Client
- /compute_ik [lab4_controller_rviz/srv/ComputeIK]
- /set_control_mode [lab4_controller_rviz/srv/SetControlMode]

Lab4_description

rviz

Publisher

Subscriber

/target [geometry_msgs/msg/PoseStamped]
/end_effector [geometry_msgs/msg/PoseStamped]
/robot_description [std_msgs/msg/String]
/workspace_points [sensor_msgs/msg/PointCloud2]

Service Server

Service Client